True/False (4×5 points)

1. If X and Y are independent random variables, their joint PMF $P(X = x, Y = y)$ will always be strictly greater than the marginal PMF $P(X = x)$.

2. For a Geometric distribution $X \sim Geom(p)$, which represents the number of trials upto & including the 1st success, the probability of requiring strictly more than 'k' trials is $(1-p)^k$.

3. For any two random events A and B (where $P(A) > 0$ and $P(B) > 0$), if knowing that B occurred increases the liklihood of A occurring (i.e. $P(A|B) > P(A)$), then knowing that A occurred must also increase the liklihood of B occuring.

4. The cumulative distribution function (CDF), $F_X(t)$, of a discrete random variable is a strictly continuous curve over all real numbers.

Fill in the blanks (5x5 points)

1. If a sample space S consists of a finite number of equally likely outcomes, the probability of an event $A \subset S$ is defined as

$P(A) =$ ______

2. Let X be the number of successes in 'n' independent Bernoulli trials with success probability $p = 0.3$. Let Y be the number of failures in those exact same 'n' trials. Then X & Y are (independent / identically distributed / both / neither) ______

3. Let X be the indicator random variable that equals 1 if event A occurs & 0 otherwise, then $E[X] =$ ______.

4. The distribution which models the number of trials upto the rth success in a sequence of independent Bernoulli trials is ______.

5. If X & Y are independent random variables and $Var(X) = a$ & $Var(Y) = b$, then

$Var(X-Y) =$ ____________ .

Free response questions:

(5 points)

① Show that $E[X] = \sum_{k=0}^{\infty} P(X>k)$ for any random variable X

s.t $\text{Range}(X) \subseteq \mathbb{Z}_{\geq 0}$

(5+10 points)

② In modern mobile gaming, players open digital "loot boxes" to obtain rare characters.

In a specific game, each loot box opened has a probability of containing a legendary character is $p = 0.1$.

(a) If a player opens boxes indefinitely, what is the expected number of boxes they must open to get their first legendary character?

Name the distribution and its parameters.

(b) To prevent extreme bad luck, the developers implement a "Pity System".

If a player opens 89 boxes in a row without getting a legendary character, the 90th box is

guaranteed to contain one (probability = 1).
let X = # boxes opened to get a legendary character under this new system.
Using ①, find $E[X]$ as a simplified expression in terms of p.

Solutions :

① Fill in the blanks

1. $\frac{|A|}{|S|}$ or (Sol.ⁿ)

writing
"number of elements in A divided by the total number of elements in S".

[redacted]

2. Sol.ⁿ: Neither [redacted]

Justification : Not independent because

[redacted]

$X+Y=n$

(knowing X, completely

[redacted] determine Y)

[redacted] Not identically distributed

$X \sim Bin(n, 0.3)$ while

$Y \sim Bin(n, 0.7)$

3. Sol.^{n}: $E[X] = P(A)$ or "probability of occurrence
[redacted] of event A".

Justification :

let $P(A)$ denote the probability of occurence of event A.

$$X = \begin{cases} 1 & \text{if } P(A) > 0 \\ 0 & \text{if } P(A) = 0 \end{cases}$$

$$E[X] = 1 \cdot P(X=1) + 0 \cdot P(X=0)$$

$$= P(A) + 0 = P(A)$$

[redacted]

4. Sol.^{n}: Negative Binomial distribution

[redacted]

5. Sol^n: $a+b$ or "$Var(X) + Var(Y)$"

[redacted]

Justification :

Sol^n 1 : [redacted] $Var(X-Y) = Var(X) + Var(-Y)$

[redacted] since if X and Y are independent, then

[redacted] X and $-Y$ are also independent.

$\Rightarrow Var(X-Y) = Var(X) + (-1)^2 Var(Y)$

[redacted]

$= Var(X) + Var(Y)$

$= a+b$

Sol^n 2 : $Var(X-Y) = E((X-Y)^2) - (E[X-Y])^2$

$= E[X^2+Y^2-2XY]$

$-(E[X]-E[Y])^2$

$= E[X^2]+E[Y^2]-2E[XY]$

$-(E[X])^2-(E[Y])^2$

$+2E[X]E[Y]$

$\Rightarrow Var(X-Y) = E[X^2]-(E[X])^2 +$

$E[Y^2]-(E[Y])^2$

$-2E[XY]+2E[X]E[Y]$

$= Var(X)+Var(Y) - 2.0$

since X & Y are independent

[redacted]

$E[XY] = E[X]E[Y]$

$= Var(X) + Var(Y) = a+b$

[redacted]

True/False :

1. Sol^n: False [redacted]

Justification:

Since X and Y are independent, then

$P(X=x, Y=y) = P(X=x)P(Y=y)$

[redacted] $\leq P(X=x)$ (since $P(Y=y) \leq 1$ for any y [redacted]

$\Rightarrow \quad P(X=x, Y=y) \leq P(X=x)$

It can never be strictly greater

[redacted]

2. Sol^n: True [redacted]

Justification :

[redacted] In order for strictly more than 'k' trials for the [redacted] 1st success → this means that the first 'k' trials must all be failures.

The probability of a single failure = $1-p$.

Since, trials are independent,

Probability of k consecutive failures $= (1-p)^k$

[redacted]

3. Sol^n: True [redacted]

Justification : Given : $P(A|B) > P(A)$

To prove : $P(B|A) > P(B)$

[redacted]

$\Rightarrow P(A|B) = \dfrac{P(A \cap B)}{P(B)}$ (Bayes rule)

[redacted]

$\Rightarrow \dfrac{P(A \cap B)}{P(B)} > P(A)$

$\Rightarrow \dfrac{P(A \cap B)}{P(A)} > P(B)$

$\Rightarrow P(B|A) > P(B)$

hence proved

4. Sol^n: False [redacted]

Justification : It is a step function

[redacted] It has "jumps" at the discrete values

It is "right continuous"

or come up with

Free response question

1. Sol.^{n}:

By definition

$$E[X] = \sum_{k=0}^{\infty} k\, P(X=k)$$

$$= \sum_{k=0}^{\infty} \left(\sum_{m=0}^{k-1} 1 \right) P(X=k)$$

$0 \leq k < \infty$

$0 \leq m \leq k-1$

$\infty > k \geq m+1$

$$= \sum_{m=0}^{\infty} \underbrace{\sum_{k=m+1}^{\infty} P(X=k)}$$

$$= \sum_{m=0}^{\infty} P(X > m)$$

[redacted]

2.(a) Solⁿ: Distribution : Geometric

Parameter : $p = 0.1$

expected value : $E[X] = \frac{1}{p} = 10$ boxes

[redacted]

(b) Sol.^{n}: Under the pity system:

For $k < 90$, $P(X > k) = (1-p)^k$

For $k \geq 90$, 90th box is guaranteed to be a success,

so it it is impossible to require more than 90 boxes to get the 1st character.

$$P(X > k) = 0 \text{ for } k \geq 90$$

Substituting this into

$$E[X] = \sum_{k=0}^{\infty} P(X > k)$$

$$= \sum_{k=0}^{89} P(X>k) + \cancel{\sum_{k \geq 90} P(X>k)}$$

$$= \sum_{k=0}^{89} (1-p)^k$$

$$= \frac{1-(1-p)^{90}}{1-(1-p)}$$

$$= \boxed{\frac{1-(1-p)^{90}}{p}}$$

[redacted]